



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

Centre for Professional and
Continuing Education

(SCTP) Advanced Professional Certificate **Data Science and AI**



Module 3 · Machine Learning & GenAI

L02

How Sure *Are We?*

Probability & Statistics for ML · Sarah's week of defending every number

Join at
slido.com
#DS6302



Where We Left Off

Priya's question is the engine of this lesson

Priya, at the end of 3.1:

"How sure are we? The model put a label on every review.

How do we know it isn't confidently wrong in ways we can't see?"

Three numbers Sarah needs to defend by Friday:

60%

of reviews are positive

→ **Distribution analysis (Tuesday)**

84%

model accuracy

→ **Confidence interval (Wednesday)**

?

coupon reduces churn

→ **A/B test + p-value (Thursday)**

What is probability theory ?

Probability theory is the mathematical framework for quantifying uncertainty. It helps us answer questions like "What are the chances?" and provides the foundation for making informed decisions when outcomes are uncertain.



Weather Prediction: Predicting the likelihood of rain.



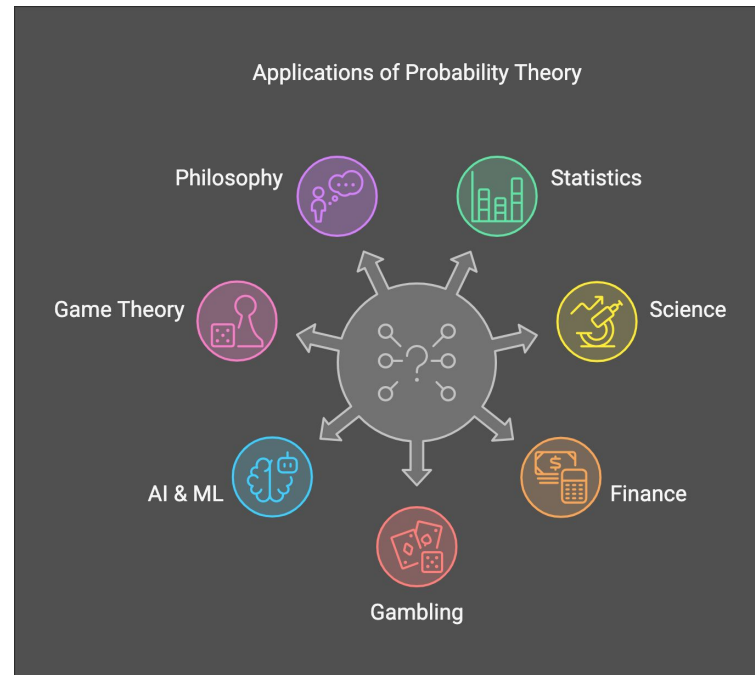
Gambling: Calculating odds in poker or roulette



Finance: Assessing potential risks and returns on investments



Medical Diagnosis: Evaluating the probability of a disease given certain symptoms



Probability Distributions



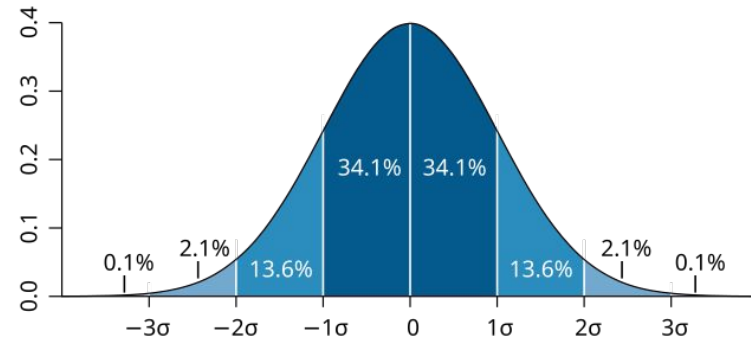
Describes how likely different outcomes of a random event to occur.

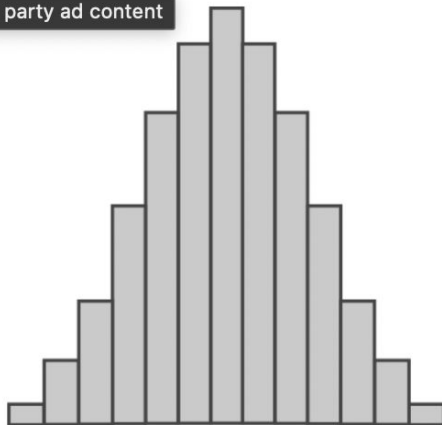


A map that shows all possible results and how often each should occur.

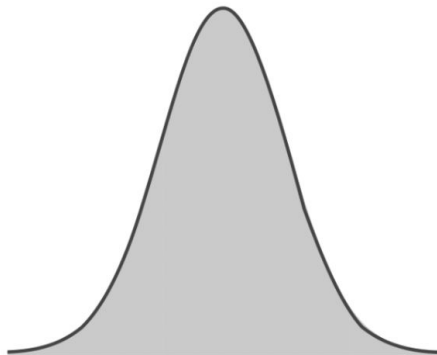
There are two main types:

Discrete and **Continuous** PDs





Discrete



Continuous

Probability Distributions: The Shape of Uncertainty

Probability distributions show us how likely different outcomes are. They're the building blocks for understanding patterns in data and making predictions about future events.

Discrete Distributions

Deal with countable outcomes like rolling dice or counting defective products. Examples include binomial and Poisson distributions.

- Binomial: success/failure scenarios
- Poisson: rare events over time

Continuous Distributions

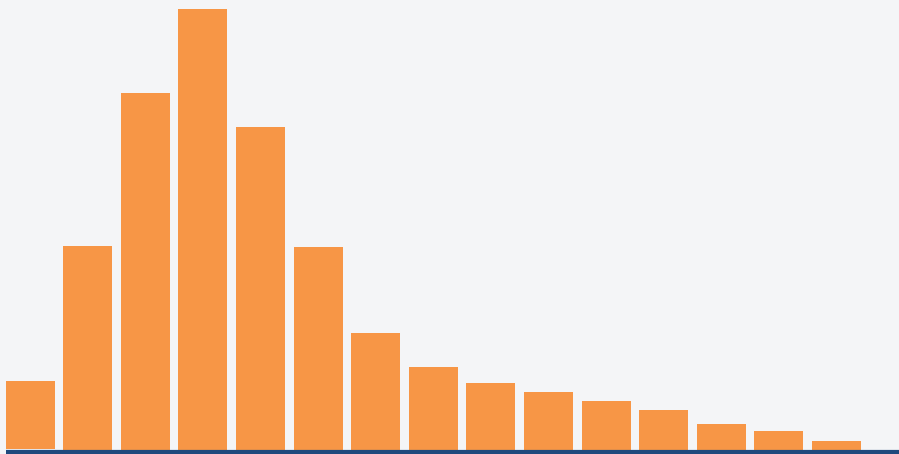
Handle measurements that can take any value within a range, like height or temperature. The normal distribution is the most famous example.

- Normal: bell-shaped, symmetric
- Uniform: equally likely outcomes

Sarah's Monday — The Polarity Histogram

Reading the shape of 10,000 reviews

What Sarah sees:



Most scores cluster in the mildly-positive range, with a long tail of very positive reviews on the right.

Sarah's first reaction:

"It's not a bell curve. There's a hump on the left, a long tail to the right."

"How do I describe this to Priya in a way that's honest?"

The vocabulary she needs:

- Skewness
- Mean vs Median
- Z-scores

Mean vs Median

The gap tells you about the skew

Worked example — UK household income (right-skewed)

£35k

Median

£52k

Mean

Three quick rules

- Mean = Median** → Data is symmetric. Quote either.
- Mean > Median** → Right-skewed. Quote MEDIAN for 'typical'; quote MEAN for 'total income generated'.
- Mean < Median** → Left-skewed. Same logic, flipped.

Z-scores — Putting Values in Context

How many standard deviations from average?

Formula

$$z = (\text{value} - \text{mean}) / \text{standard deviation}$$

Worked example — Sarah's exam score

Sarah scored 85 on a test. Is that good?

If class average = 60, std = 10 $z = (85-60)/10 = 2.5$ → Top ~1% of the class

If class average = 82, std = 2 $z = (85-82)/2 = 1.5$ → Above average, not exceptional

Same raw score (85) means very different things in different contexts. Z-scores reveal the difference.

CODE-ALONG

Open: **02_distributions.ipynb**

What you'll do

1. Plot Sarah's polarity & review-length distributions side-by-side.
2. Compute the Measures of Central Tendency (Mean, Median, and Mode) for these distributions.
3. Spot which summary statistic to quote for which question based on the skew.



30 minutes · Stop at the  Extension marker · Extension is self-paced after class

Sarah's Wednesday — 84% From 200 Reviews

Is it 'exactly' 84%? Or is that just one sample's answer?

Sarah's plan to validate the L01 sentiment model:

Hand-label 200 reviews herself. Compare to the model's predictions. Report agreement rate.

The result

167 of 200 reviews agreed.

That's 83.5% — rounded to:

84%

Priya's follow-up

"So the model is EXACTLY 84% accurate?"

Sarah pauses. If she had labelled a different 200 reviews, would she get the same number?

Probably not. But how close?

Law of Large Numbers

"As the number of trials increases, the observed probability converges to the true probability"

Example:

As the number of coin flips increases, the observed probability of heads will converge to 50%.

- Foundation for probability and statistics.
- Relationship between **theoretical** probability and **empirical** results.



The Central Limit Theorem (CLT)

Why Sarah's histogram looks bell-shaped

The Central Limit Theorem, in plain English

*"Even if the underlying data isn't normally distributed, the **AVERAGE** of many samples from that data will follow a normal distribution. The larger each sample, the tighter the normal becomes."*

Three things this lets you do

- 1 Compute confidence intervals on almost any accuracy metric**
(model accuracy, conversion rate, click-through rate, complaint rate)
- 2 Run hypothesis tests without assuming the raw data is normal**
(t-test, z-test, A/B tests — all rely on the CLT)
- 3 Reason about sample size**
(want a tighter estimate? Increase n . The bell curve gets narrower.)

Confidence Intervals

The bracket around every accuracy number

Plain English

*"A 95% confidence interval is a range that captures the true value about 95% of the time, **IF** we were to repeat the experiment many, many times."*

Formula for a proportion (like accuracy)

$$\text{Lower} = \hat{p} - 1.96 \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \qquad \text{Upper} = \hat{p} + 1.96 \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Sarah's 84% from 200 reviews

$$\hat{p} = 0.84, \quad n = 200 \quad \rightarrow \quad \text{CI} = [78.4\%, 89.6\%]$$

Thursday — Aisha's Apology Coupon Idea

Will it reduce complaints — or is it expensive noise?

Aisha · Customer Service

Send a 10% apology coupon to every NEGATIVE-tagged customer.
She believes it'll turn some unhappy customers around.

Marcus · CTO

Thousands of pounds a week + engineering work to wire the trigger.
He wants evidence before signing off.

Sarah · Data Analyst

Test it properly. Define the hypothesis. Run an A/B test. Compute a p-value.
Her L02 toolkit makes her the right person to answer.

The P-value

The probability that the data is just noise

Plain English

" $P = P(\text{data this extreme} \mid \text{null hypothesis is true})$. If p is small, the data would be unlikely under H_0 , so we reject H_0 ."

Standard decision rule ($\alpha = 0.05$):

$p < 0.05$

Reject the null hypothesis.

The observed difference is unlikely under H_0 .
There's evidence of a real effect.

$p \geq 0.05$

FAIL to reject the null hypothesis.

Not enough evidence to declare an effect.
(NOT the same as 'no effect' — see next slide.)

Friday — The Three Defended Numbers

Same numbers, now backed by the right tools

60%

of reviews are positive

84%

model accuracy

~6pp

drop in 30-day complaint rate from
the coupon ($p < 0.05$)

Distribution analysis ·
Median + chart of skew

Confidence interval ·
95% CI: [78%, 89%] from $n=200$

A/B test · Two-proportion z-test

Priya: *'Looks great. But you used a pre-trained model. Can you build one TRAINED on NorthStar's own data?'*

Bridge to L03

Marcus, after the Friday meeting:

*"OK. The sentiment model holds up. But you used a pre-trained one — off the shelf.
Can you build us a model that predicts CHURN from our own customer data?
Something trained on NorthStar behaviour, not some generic internet corpus?"*

Lesson 3 — Supervised Learning Foundations

Train/test split · preprocessing · regression vs classification · linear & logistic models.

First time you train a model end-to-end on NorthStar's own data.

Assignment due before L03 — same workflow, new dataset (Lakeside Bank).

End of Lesson - Exit Ticket

Survey Link

Slido link will be provided in class

